

Liver Lesion Diagnosis Challenge on Multi-phase MRI:

Structured description of the challenge design

CHALLENGE ORGANIZATION

Title

Use the title to convey the essential information on the challenge mission.

Liver Lesion Diagnosis Challenge on Multi-phase MRI

Challenge acronym

Preferable, provide a short acronym of the challenge (if any).

LLD-MMRI2023

Challenge abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Liver cancer remains one of the most severe diseases threatening human health globally. Early diagnosis plays an important role in improving the survival rate and reducing patient suffering. In clinical practice, multi-phase contrast-enhanced magnetic resonance imaging (MRI) can deliver more accurate liver lesion diagnosis, as there is more detailed visual information such as clearer blood vessels, mediastinum, and different image enhancement manners of various lesions. However, liver lesion diagnosis in 3D multi-phase images is a challenging task since liver lesion has variant shape, diverse size, irregular boundaries, and low signal-to-noise ratio. Meanwhile, each phase exhibits different image properties such as intensity and contrast, that is, each phase reflects different lesion characteristics. Moreover, to the best of our knowledge, there is no publicly available dataset of liver lesions with multi-phase MR scans, which limits the development of automatic computer-aided diagnosis (CAD) systems for multi-phase liver lesion identification.

We plan to host the first Liver Lesion Diagnosis Challenge on Multi-phase MRI (termed LLD- MMRI2023) to promote the research of such CAD systems. In our challenge, 498 multi-phase MRI lesion cases are publicly released, which are carefully annotated by 3 experienced radiologists. Each case comprises 8 different phases, including the non-contrast phase, arterial phase, venous phase, delay phase, T2-weighted imaging phase, diffusion-weighted imaging phase, T1 in-phase, and T1 out-of-phase. We assigned 316 cases for training, 78 cases for validation, and 104 cases for testing. The classification performance is reported with F1-score and Cohen's Kappa coefficient.

Challenge keywords

List the primary keywords that characterize the challenge.

Liver Lesion, Multi-phase MRI, Diagnosis, Classification

Year

The challenge will take place in ...

2023

FURTHER INFORMATION FOR MICCAI ORGANIZERS

Workshop

If the challenge is part of a workshop, please indicate the workshop.

None

Duration

How long does the challenge take?

Half day.

Expected number of participants

Please explain the basis of your estimate (e.g. numbers from previous challenges) and/or provide a list of potential participants and indicate if they have already confirmed their willingness to contribute.

We believe this challenge will attract many participants. In addition, we will set up a prize pool (\$5,000-\$10,000) to reward the top-5 teams, including one first-place prize, one second-place prize, and one to three third-place prizes, hoping to attract more participants to actively participate. We expect 100-200 participants.

Publication and future plans

Please indicate if you plan to coordinate a publication of the challenge results.

We plan to publish at least one paper in medical image analysis journals (MIA, JBHI, and Nature Partner Journal Digital Medicine, etc.) summarizing the challenging results and discussing future research directions. The top-5 teams can have up to 5 co-authors each, while the 6th to 10th place teams can have up to 3 co-authors each. In the paper, we will publish a substantial summary of the challenge results.

Space and hardware requirements

Organizers of on-site challenges must provide a fair computing environment for all participants. For instance, algorithms should run on the same computing platform provided to all.

LLD-MMRI2023 is an off-site challenge. Participants are required to independently train and validate their models on their own computing infrastructure. For testing, the website <https://github.com/LMMMEng/LLD-MMRI2023> will be used to present the challenge details and release the dataset, ranking method, code for baseline, and evaluation.

TASK: Classification

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Same as above

Keywords

List the primary keywords that characterize the task.

Liver Lesion, Multi-phase MRI, Classification

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Meng Lou (Artificial Intelligence Laboratory, Deepwise Healthcare, Beijing, China, E-mail:
loumeng@deepwise.com)

Xiaoqing Liu (Artificial Intelligence Laboratory, Deepwise Healthcare, Beijing, China, E-mail:
liuxiaoqing@deepwise.com)

Yuqing Zhang (Department of Radiology, Ningbo Medical Center Lihuili Hospital, Ningbo, Zhejiang, China, E-mail:
zyqlucky168@163.com)

Yizhou Yu (Department of Computer Science, The University of Hong Kong, Pokfulam, Hong Kong, E-mail:
yizhouy@acm.org)

Hong-Yu Zhou (Department of Computer Science, The University of Hong Kong, Pokfulam, Hong Kong, E-mail:
whuzhouhongyu@gmail.com)

b) Provide information on the primary contact person.

Yizhou Yu (Department of Computer Science, The University of Hong Kong, Pokfulam, Hong Kong, E-mail:
yizhouy@acm.org)

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place. Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

Open call challenge.

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

MICCAI.

b) Report the platform (e.g. grand-challenge.org) used to run the challenge.

<https://github.com/LMMMEng/LLD-MMRI2023>

c) Provide the URL for the challenge website (if any).

<https://github.com/LMMMEng/LLD-MMRI2023>

Participation policies

a) Define the allowed user interaction of the algorithms assessed (e.g. only (semi-) automatic methods allowed).

Fully automatic.

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or to publicly available data including (open) pre-trained nets.

Publicly available data is allowed.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard.

d) Define the award policy. In particular, provide details with respect to challenge prizes.

We will award prizes of \$5,000-\$10,000 to the top-5 finishers according to the rankings, including one first-place prize, one second-place prize, and one to three third-place prizes.

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

1) All the results will be publicly displayed on the leaderboard.

2) The top-5 teams on board will be invited to give a 5-10 minute presentation for the MICCAI2023 challenge session.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

1) The challenge organizers will publish at least one paper in the aforementioned journals. The top-5 teams can have up to 5 co-authors each, while the 6th to 10th teams can have up to 3 co-authors each. Note that the paper will be published on arXiv first.

2) All participating teams are encouraged to publish their results individually after the challenge, but they should cite the arXiv paper mentioned above.

3) The embargo period ends at the MICCAI2023 conference (Oct. 8, 2023).

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

1) GitHub Repository: data download links will be available in our GitHub repository, and we will also provide a code repository comprising data processing and loading, model training and evaluation, prediction, and etc.

Meanwhile, a baseline model will also be provided for reference. More details can be found at <https://github.com/LMMMEng/LLD-MMRI2023>.

2) Data Release Stage: The training set, validation set, and test set will be released sequentially in three stages. First, the training dataset (with annotations) and the validation dataset (without annotations) will be released on April 28th. Second, annotations on the validation set will be accessible on July 8th. Third, the test dataset (without annotations) will be released on August 2nd. That is, participants will be provided with annotated training and validation datasets as well as non-annotated test data before the final evaluation.

3) Algorithm Development Stage: participants need to design and output a satisfactory model based on the released datasets and annotations on their own computing devices. When the validations/test sets are released, a prediction code will be provided to generate the prediction score for each case from the test set for reference. It should be noted that participants must generate prediction results in strict accordance with our pre-defined file format (case id with predictions for each category).

4) Algorithm Evaluation Stage: participants are required to submit their predictions on the test set within a specified time frame. We will then calculate performance metrics (Macro-averaged F1-score and Cohen's Kappa coefficient) of their algorithms based on their submissions and ground-truth annotations. The challenge consists of four evaluation stages. In the first three stages, the ranking will be updated based on the predicted results of the algorithm on the validation set. Participants will have a 24-hour submission window on May 26th, June 16th, and July 7th to submit their prediction results. On July 8th, annotations on the validation set will be released to further support model design and training. In the final stage, the test set (without annotations) will be released on August 2nd. Participants are required to submit their predictions on August 4th, and the predicted results at this stage will determine the final ranking. Regarding the submission method, participants should send their prediction files to the official email address specified on GitHub within a specified period after completing predictions. Note that if a team commits multiple times within a single submission window, we will only use the last successful commit. Then, the algorithm performance will be shown based on the submitted results and ground-truth annotations of the test set.

5) Result Display Stage: We will sort out the effective results in a short time and display the scores and rankings of each team on GitHub.

6) Verification Stage: To ensure the fairness of the challenge, the top-10 teams will be required to disclose their codes and model weights on GitHub or other publicly accessible websites for verification. We will use these codes and model weights to verify that the reproduced results are consistent with the submitted predictions. If teams fail to disclose their codes and model weights within the stipulated time frame, they will be removed from the leaderboard. If serious discrepancies are detected using disclosed codes and model weights, we will notify the corresponding teams to take remedial actions. If they fail to complete within the allotted time, they will be

removed from the leaderboard and the leaderboard will be adjusted accordingly. New teams that subsequently enter the top-10 will also be required to comply with the same rules.

More detailed submission instructions will be given at <https://github.com/LMMMEng/LLC-3DAM-MRI2023>.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

The challenge consists of four evaluation stages. In the first three stages, the ranking will be updated based on the predicted results of the algorithm on the validation set. Participants will have a 24-hour submission window on May 26th, June 16th, and July 7th to submit their prediction results. On July 8th, annotations on the validation set will be released to further support model design and training. In the final stage, the test set (without annotations) will be released on August 2nd. Participants are required to submit their predictions on August 4th, and the predicted results at this stage will determine the final ranking.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

Registration: March 28, 2023;

Dataset release: The training set, validation set, and test set will be released sequentially in three stages. First, the training dataset (with annotations) and the validation dataset (without annotations) will be released on April 28th. Second, annotations on the validation set will be accessible on July 8th. Third, the test dataset (without annotations) will be released on August 2nd.

Submission deadline: August 4, 2023;

Winner and invitation speakers: Oct 8, 2023--Oct 12, 2023.

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

The data used in this challenge were collected retrospectively from the Ningbo Medical Center Lihuili Hospital, and its use was approved by the Institutional Review Board of the Ningbo Medical Center Lihuili Hospital. The ethics approval number: KY2022SL446-01

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

CC BY NC ND.

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

The codes to produce the baseline results, evaluations, rankings, and other relevant documentation will be available on our GitHub repository <https://github.com/LMMMEng/LLD-MMRI2023>, once our proposal is accepted. Evaluations will be posted to contestants prior to open validation leaderboard submissions. In this way, all participants will verify the evaluation code by validation dataset, ensuring the fairness of the challenge.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

All teams are encouraged to share their code but are not obliged to do so.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

- 1) Deepwise Healthcare and The University of Hong Kong will provide funding for the challenge.
- 2) Only challenge organizers have access to the labels of the test dataset.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Diagnosis, Research.

Task category(ies)

State the task category(ies).

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation
- Tracking

Classification.

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort consists of patients with early diagnosis of focal liver lesions.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

Patient inclusion and exclusion criteria have been added to the revised proposal as follows:

Inclusion criteria: patients who underwent multi-phase MRI scans and had at least one focal liver lesion.

Exclusion criteria: (1) patients with any treatment for focal liver lesions before MR scanning, (2) patients with poor MR imaging quality, and (3) patients diagnosed with diffuse liver disease.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

Multi-phase Magnetic Resonance Imaging (MRI)

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

The acquired full volume data are provided together with the coordinates of the lesion bounding box and its corresponding lesion type.

b) ... to the patient in general (e.g. sex, medical history).

No patient information.

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

Liver shown in abdominal multi-phase MRI data.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

Differentiate 7 types of liver lesions including 4 benign types (Hepatic hemangioma, Hepatic abscess, Hepatic cysts, and Focal nodular hyperplasia) and 3 malignant types (Intrahepatic cholangiocarcinoma, Liver metastases, and Hepatocellular carcinoma).

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Macro-averaged F1-score and Cohen's Kappa coefficient

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

Multi-phase MRI data acquired from 1.5T or 3.0T MRI systems manufactured by Kangda, GE, and Philips.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

The image acquisition protocol is as follows:

Non-contrast scan (coronal plane): TR=953ms, TE=80ms, FOV=370mm, matrix=256*237, slice thickness=5mm, and slice spacing=1mm.

T2WI: TR=1918ms, TE=80ms, FOV=370mm, matrix=276*276, slice thickness=5mm, and slice spacing=1mm.

DWI: TR=1856ms, TE=63ms, FOV=370mm, matrix=132*116, slice thickness=5mm, and slice spacing=1mm.

T1WI out-of-phase/in-phase: TR=3.8ms, TE=1.35ms, FOV=370mm, matrix=276*236, slice thickness=4 mm, and slice spacing=2mm, and breath holding time=10-15s.

Contrast-enhanced scans were performed based on the injection of 0.2 mmol/kg of GD-DTPA through the cubital vein (2.0mL/s). The arterial phase was when the contrast agent entered the left ventricle, the portal phase was after 55s, the venous phase was after 90s, and the delay phase was after

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

Ningbo Medical Center Lihuili Hospital, Ningbo, Zhejiang, China

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

Two radiologists with 6 and 8 years of experience respectively were responsible for annotating the dataset and another senior radiologist with more than 10 years of experience was responsible for annotation confirmation and/or correction (if necessary).

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

A case refers to all information available for one liver lesion. This information includes (1) multi-parametric multiphase MRI scans (including non-contrast phase, arterial phase, venous phase, delay phase, T2- weighted imaging phase, diffusion-weighted imaging phase, T1 in phase, and T1 out of phase) and (2) annotation information (including the bounding box of the lesion and its corresponding type in each phase).

b) State the total number of training, validation and test cases.

Train set: {'Hepatic hemangioma': 50, 'Intrahepatic cholangiocarcinoma': 37, 'Hepatic abscess': 34, 'Hepatic metastasis': 32, 'Hepatic cyst': 34, 'Focal nodular hyperplasia': 29, 'Hepatocellular carcinoma': 100, 'All': 316}

Validation set: {'Hepatic hemangioma': 13, 'Intrahepatic cholangiocarcinoma': 9, 'Hepatic abscess': 8, 'Hepatic metastasis': 8, 'Hepatic cyst': 8, 'Focal nodular hyperplasia': 7, 'Hepatocellular carcinoma': 25, 'All': 78}

Test set: {'Hepatic hemangioma': 16, 'Intrahepatic cholangiocarcinoma': 12, 'Hepatic abscess': 12, 'Hepatic metastasis': 11, 'Hepatic cyst': 11, 'Focal nodular hyperplasia': 10, 'Hepatocellular carcinoma': 32, 'All': 104}

Total: {'Hepatic hemangioma': 79, 'Intrahepatic cholangiocarcinoma': 58, 'Hepatic abscess': 54, 'Hepatic metastasis': 51, 'Hepatic cyst': 53, 'Focal nodular hyperplasia': 46, 'Hepatocellular carcinoma': 157, 'All': 498}

c) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

Training cases: 316 (A relatively large amount of data used to train a robust model).

Validation cases: 78 (A reasonable amount of data used for participants to validate the evaluation code and tune hyper-parameters).

Test cases: 104 (A reasonable amount of data for a fair final leaderboard).

d) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The training, validation, and test sets have equal class distribution.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

Two radiologists with 6 and 8 years of experience respectively were responsible for annotating the dataset, where each image was annotated by one of the two radiologists and another senior radiologist with more than 10 years of experience confirmed and/or corrected (if necessary) the annotation. The annotation and review process was completed using an annotation system developed by Deepwise Healthcare. Pathology reports are the reference standard for annotating bounding boxes and corresponding lesion types on multi-phase MRI images.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

Prior to the annotation process, annotators are trained to draw the bounding box of each lesion and identify the corresponding lesion type strictly following the description of the pathology report.

Step 1: Three experienced radiologists select multi-phase MRI sequences based on available pathology reports. After filtering out sequences containing low-quality images, the remaining MRI sequences constitute an unannotated multi-phase MRI dataset covering 7 different types of liver lesions.

Step 2: Upload the above-mentioned dataset to the annotation system of Deepwise Healthcare. This annotation system has an easy-to-use labeling tool that provides an interactive user interface for the user to draw a rectangular bounding box around the lesion on the image of each phase and select its corresponding lesion type.

Step 3: First, annotators need to refer to the pathology report to determine the number of lesions and the corresponding lesion types in the multi-phase MRI images. Then, they combine pathological information with image characteristics to draw a bounding box for each lesion. Each bounding box is required to be as close as possible to the boundary of the lesion. Finally, annotators are required to select a lesion type for each lesion strictly adhering to the lesion diagnosis in the pathology report. There are 4 benign types (Hepatic hemangioma, Hepatic abscess, Hepatic cysts, and Focal nodular hyperplasia) and 3 malignant types (Intrahepatic cholangiocarcinoma, Liver metastases, and Hepatocellular carcinoma).

Step 4: Once the annotators submit their annotations, another senior radiologist acts as an approver to review the annotation results, along with the corresponding MRI sequences and pathology reports. If the generated annotations are unsatisfactory, the approver will carefully correct them until satisfactory quality is achieved. Otherwise, annotations are labeled as the final ground-truth for their associated lesions.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

Annotators: Two radiologists with 6 years and 8 years of experience, respectively.

Approver: A senior radiologist with more than 10 years of experience.

Annotation standard: Pathology Report

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Each lesion was annotated by one of the two radiologists and reviewed by another more experienced senior radiologist. If the annotation is of satisfactory quality (according to the senior radiologist), it will be labeled as ground-truth for the lesion. Otherwise, the senior radiologist will correct the annotation.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

- 1) Remove the private information and convert DICOM files into nii.gz files.
- 2) Manually review each image for quality control.
- 3) The lesion region was cropped from each phase.

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

- 1) For invasive lesions with blurred boundaries, their bounding boxes may not accurately cover the lesions.
- 2) When multiple lesions coexist closely in a patient, the radiologists may not be able to accurately match these lesions on the multi-phase MRI scans from the pathology report, but this scenario is extremely rare.

b) In an analogous manner, describe and quantify other relevant sources of error.

Another potential source of error source is image artifacts in MRI, which can affect image quality.

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

Macro-averaged F1-score and Kappa coefficient

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

- 1) F1-score takes both recall and precision into consideration, which is calculated as:

$$\text{F1-score} = (2 * \text{recall} * \text{precision}) / (\text{recall} + \text{precision})$$

2) Cohen's kappa, a quantitative measure of the reliability for two raters evaluating the same thing, is calculated as follows:

$$\text{Kappa coefficient} = (\text{Po} - \text{Pe}) / (1 - \text{Pe})$$

where Po is the observed relative agreement between raters and Pe is the hypothetical probability of agreement by chance.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking.

1) Calculate Cohen's Kappa coefficient.

2) Calculate the F1-score for each lesion type, and then calculate the macro-average of F1-score (m-F1) for all lesion types.

3) Rank the average score of m-F1 and Cohen's Kappa coefficient as the final task score.

4) Tie if the scores are equal.

b) Describe the method(s) used to manage submissions with missing results on test cases.

If an algorithm fails to generate an outcome metric for a particular test case, we will penalize the final results to ensure fairness. To be more specific, a penalty value will be subtracted from the evaluation score, and its calculation formula is as follows: $\text{evaluation score} * [(\text{number of missing cases} * 4) / (\text{total number of test cases})]$.

c) Justify why the described ranking scheme(s) was/were used.

Calculating the average of the two metrics can reflect the performance of the algorithm more comprehensively and avoid the problem that the algorithm performs well on one metric but poorly on others. Specifically, for any lesion type, the F1-score is computed from the confusion matrix, which maintains a balance between precision and recall to handle uneven class distribution. The Cohen's Kappa coefficient can reflect the consistency between predicted and real results, which can handle unbalanced classification distribution.

Statistical analyses

a) Provide details for the statistical methods used in the scope of the challenge analysis. This may include

- description of the missing data handling,
- details about the assessment of variability of rankings,
- description of any method used to assess whether the data met the assumptions, required for the particular statistical approach, or
- indication of any software product that was used for all data analysis methods.

We will use box plots to assess the variability in the classification scores (i.e., averages of AUC and F1 scores) across different teams on different lesion types. A box plot is a graphical representation of data that shows the median, the upper and lower quartiles, and minimum and maximum values. It allows us to see at a glance how the rankings are spread out and whether there are any outliers that are significantly different from the rest of the data. In general, more compact box plots indicate more consistent and less variable performance across all teams on a given lesion type. In contrast, the wider box plots show that more erratic and inconsistent performance across different teams.

Moreover, we will also a confusion matrix analysis to more clearly show the details of each predicted class. A

confusion matrix is a table containing the distribution of classifier performance over the data. Assuming there are N classes, the confusion matrix is an $N \times N$ matrix used to evaluate the performance of a classification model. It shows how well the model is performing, where it needs improvement, and what mistakes it made.

b) Justify why the described statistical method(s) was/were used.

Box plots are useful for data distribution comparison, by allowing you to see the shape, center, and spread of the data at a glance. They are especially useful for comparing the same dataset across different groups. Additionally, a confusion matrix can reveal which sub-types are commonly confused with which other sub-types.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

Combining algorithms via ensembling, inter-algorithm variability, common problems/biases of the submitted methods, or ranking variability will be analyzed in the future for publication in top journals in the field of medical image analysis.

ADDITIONAL POINTS

References

Please include any reference important for the challenge design, for example publications on the data, the annotation process or the chosen metrics as well as DOIs referring to data or code.